Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

mkind, Munich

Robotics Systems Engineer

Hi mkind team,

I'd like to apply for the Robotics Systems Engineer role. I like that you want people who build things with their hands and take on more ownership over time, because that is how I have worked so far.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton: a variable stiffness mechanism, with a worm gear pretensioning the spring, working in parallel with a 6 Nm motor. I optimised it with a genetic algorithm, modelled it in SolidWorks, 3D printed the first parts and had the final ones manufactured. I integrated the motor, encoder and force/torque sensors and measured its stiffness on a physical setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I have also built data collection setups with real people, which sounds close to what you do in your lab. I designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms and used it to quantify human arm impedance with 6 participants. Qualisys cameras tracked the arm, and I implemented an Arduino-based trigger so the robot's motion and the recording stayed in sync. I also collected EMG data from 12 participants with the Qualisys system.

In my Master's at CSIR-CMERI I fabricated pneumatic artificial muscles for a soft exosuit and designed the outer PID loop of a neural network-based gain-scheduled controller, with integral anti-windup and error nonlinearisation, in MATLAB/Simulink and ran it in LabVIEW. As a student I built an arm exoskeleton, an exo glove, a Go-Kart and a hybrid Efficycle. I also paint and do graphic design (Senior Diploma in Fine Arts; Illustrator, Photoshop, Procreate), which helps me sketch and explain ideas quickly.

I'm ready to move to Munich and I'm willing to learn German. I would love to bring new and innovative ideas to your robot cells, and to keep improving them from what we learn in the lab and at customer sites. My portfolio is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder